

Week Four Technical Report

Backend Development with Express.js and Microsoft SQL Server

Trainee: David Karanja

Trainer: Brian Kemboi

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Teach2Give QA & QE Training Program

Overview

This week focused on backend development using Express.js and connecting an Express server to a Microsoft SQL Server database. We set up the Node.js environment, configured TypeScript with Express, and organised our server code using a modular layered architecture.

Training Modules Covered

We started by preparing the development environment: installing Node.js, setting up TypeScript, and adding required packages with pnpm. For packages labelled with DT, we installed the corresponding devDependencies so TypeScript could work with them.

We then connected the Express server to the MSSQL database and performed CRUD operations. The project structure was separated into six layers: client (Postman), router, controller, service, repository, and database. We practised both an initial approach where the controller handled most tasks and the recommended layered approach where each layer handled its responsibilities. The week ended with authentication and tokenization using JWT and password hashing with bcrypt.

Practical Exercise and Project

For the group assessment we built a Car Management System API. We created a CarManagementDB in SQL Server with tables for Car, Customer, Booking, and Location, and seeded each table with test data. Using TypeScript and Express, we implemented CRUD endpoints for all entities, added JWT-based authentication, used bcrypt for password hashing, managed configuration with a .env file, and tested endpoints using Postman. We documented the API and the Postman collection as part of the deliverables.

Challenges Faced

One of the main challenges I encountered this week was connecting the Express server to the Microsoft SQL Server database. I had forgotten my database password, which initially prevented me from establishing a connection. However, through research and guidance,

I learned how to reset and recover access to the database. This process not only resolved the issue but also helped me understand more about SQL Server authentication and configuration.

Recommendations

I recommend using version control (GitHub) frequently to avoid conflicts and to back up work. Practising requests in Postman to understand how endpoints behave. Reviewing authentication steps early to help reduce setup problems later. Additionally, I'd say its important to keep database credentials secure but accessible and document connection procedures properly to avoid delays caused by forgotten passwords or misconfigurations.

Conclusion

Week 4 gave practical experience in building and securing a backend API. I learned how to connect Express to a real database, organise server code into layers, and protect routes with JWT. Working on the group project improved my ability to collaborate and to apply the concepts covered during the week.

Signature:

David Karanja